

WEEK 4 - ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

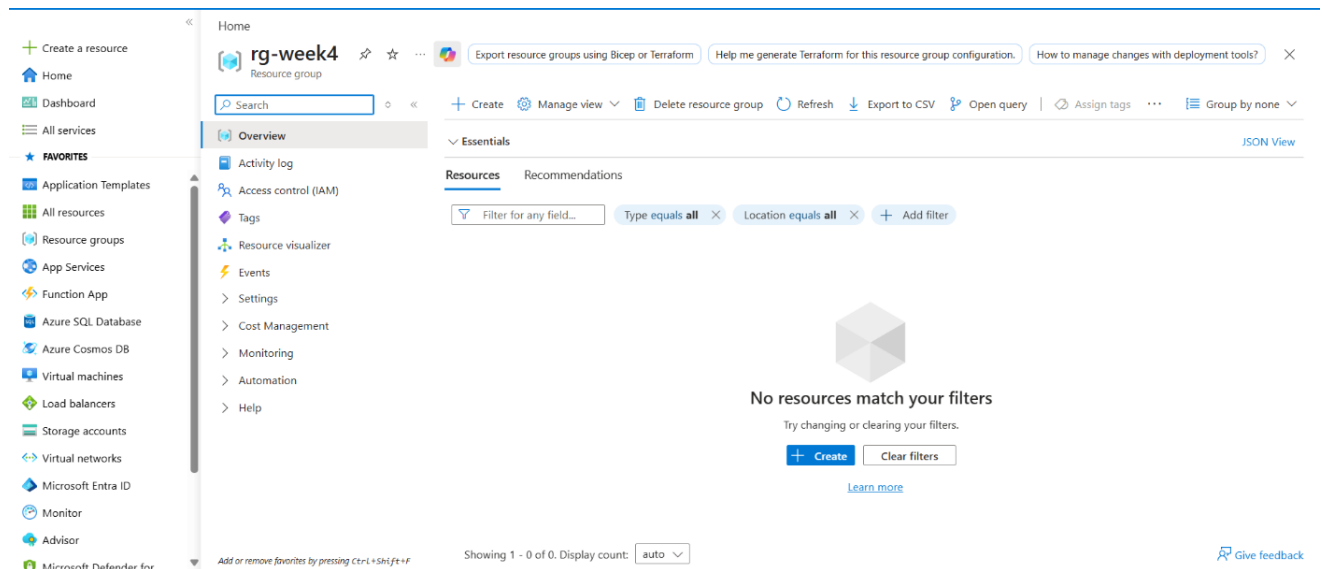
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



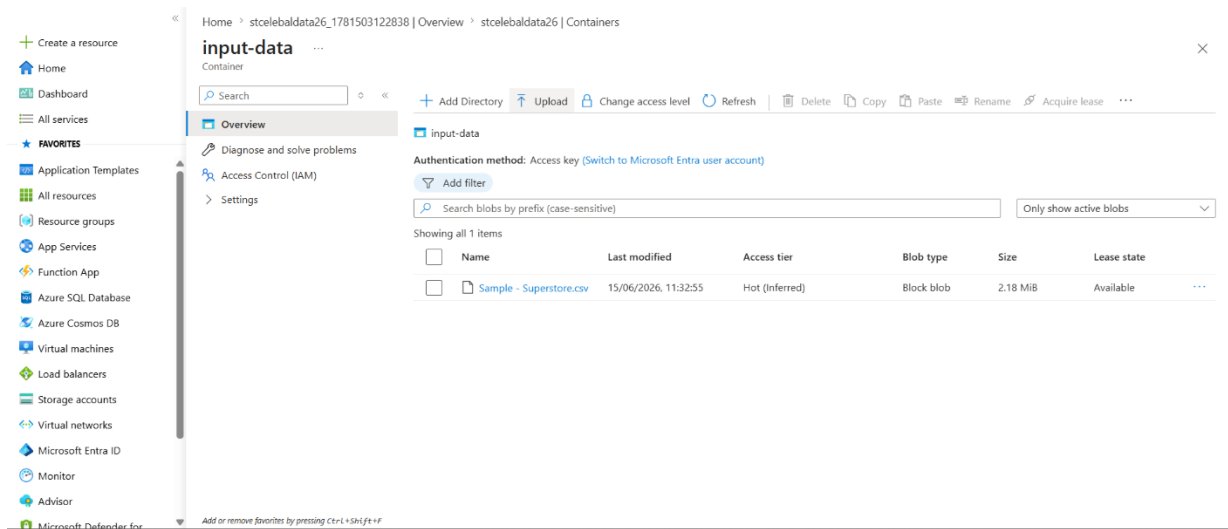
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

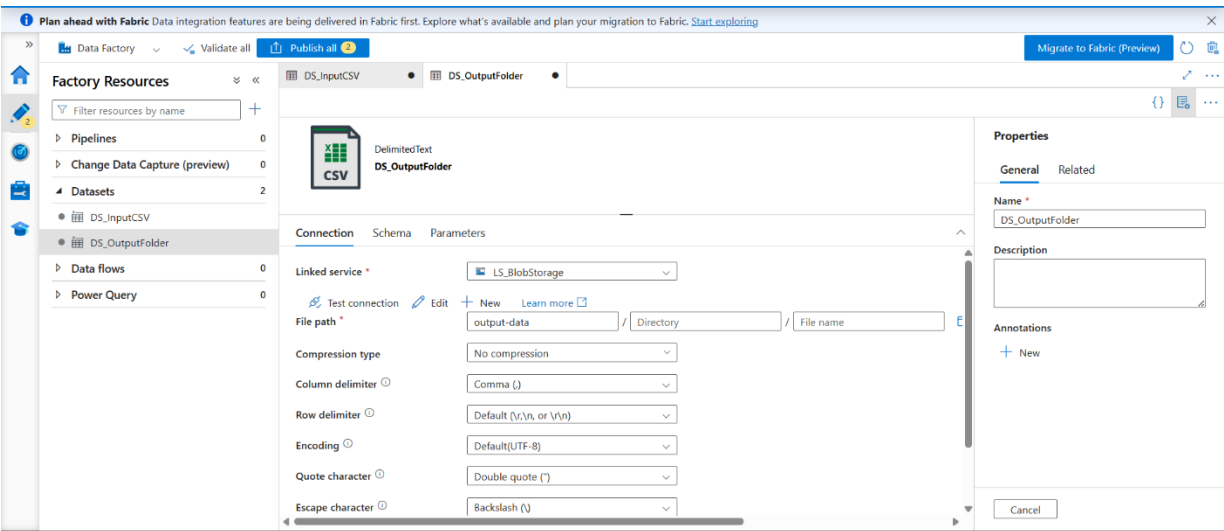
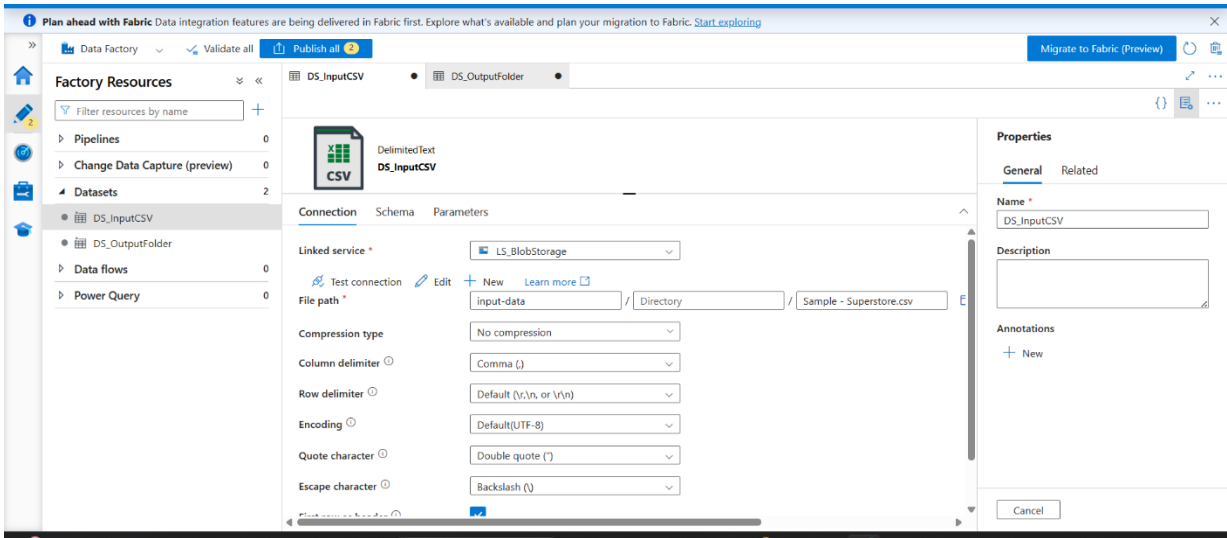
- Screenshot of Linked Service

The screenshot shows the Azure portal interface for a deployment. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area is titled 'Home' and 'CreateDataFactory-20260615113509 | Overview'. It features a search bar, action buttons (Delete, Cancel, Redeploy, Download, Refresh), and a list of tabs (Overview, Inputs, Outputs, Template). The 'Overview' tab is active, displaying a green checkmark and the message 'Your deployment is complete'. Below this, deployment details are shown: Deployment name: CreateDataFactory-202606151135..., Start time: 15/06/2026, 11:36:35, Subscription: Azure subscription 1, Correlation ID: b573cd4e-dc27-4289-b67c-d6935..., and Resource group: rg-week4. A 'Go to resource' button is present. The right sidebar contains several recommendations: 'Cost management' (Get notified to stay within your budget...), 'Microsoft Defender for Cloud' (Secure your apps and infrastructure...), 'Free Microsoft tutorials' (Start learning today...), and 'Work with an expert' (Azure experts are service provider partners...).

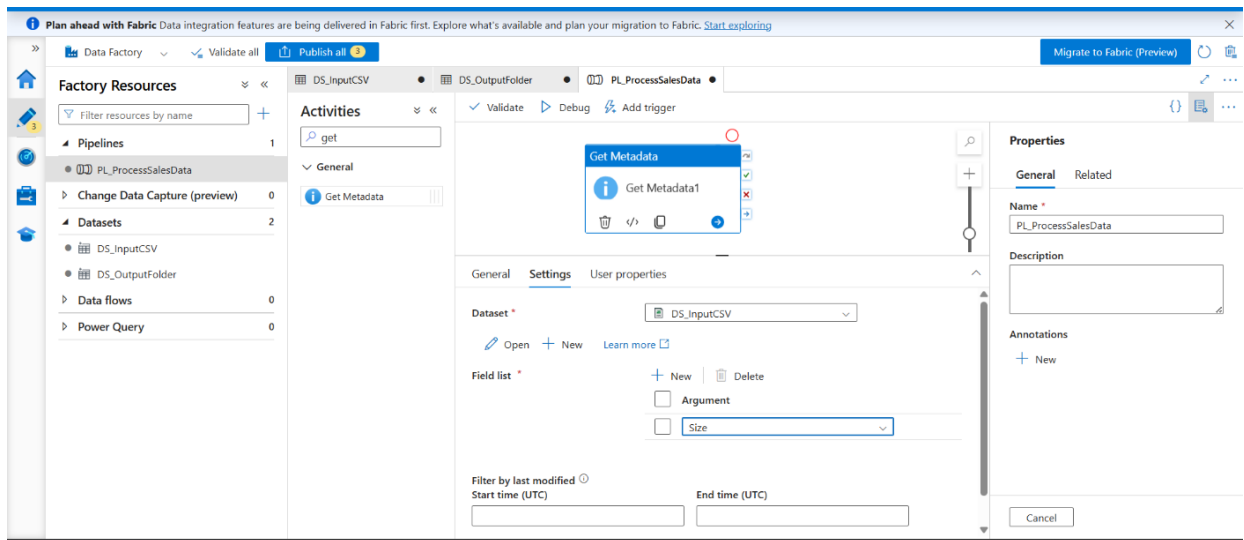
The screenshot shows the 'Linked services' page in the Azure portal. The left sidebar contains navigation links for 'Data Factory', 'Validate all', 'Publish all', and 'Migrate to Fabric (Preview)'. The main content area is titled 'Linked services' and includes a 'New' button, a 'Filter by name' input, and an 'Annotations: Any' dropdown. Below this, a table displays the linked services. The table has columns for 'Name', 'Type', 'Related', and 'Annotations'. There is one item in the table: 'LS_BlobStorage' of type 'Azure Blob Storage' with a 'Related' value of '0'.

Name	Type	Related	Annotations
LS_BlobStorage	Azure Blob Storage	0	

- Screenshot of dataset



- Screenshot of Get Metadata activity



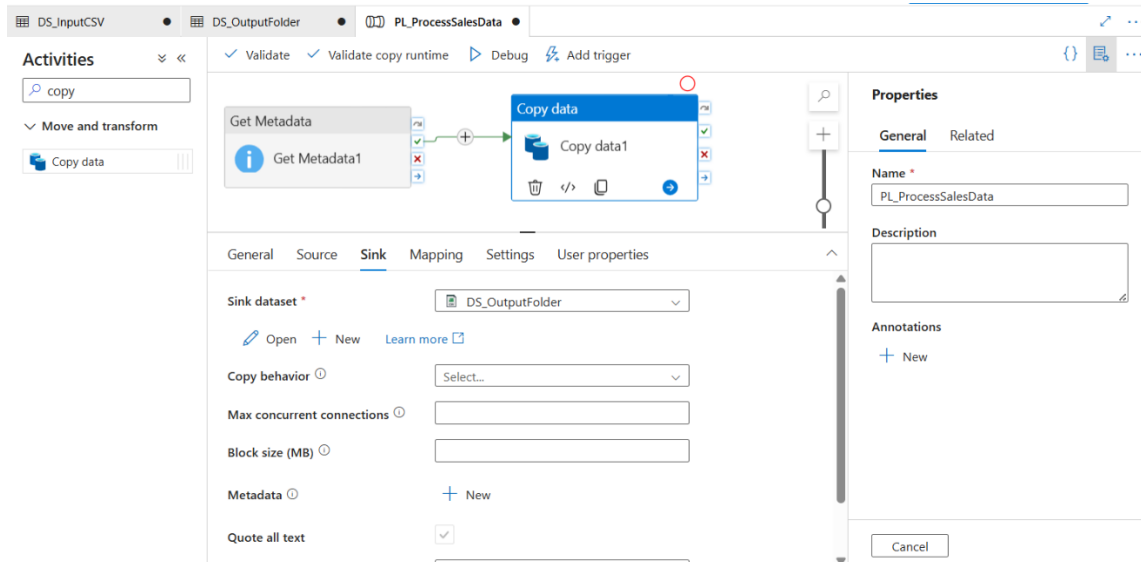
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



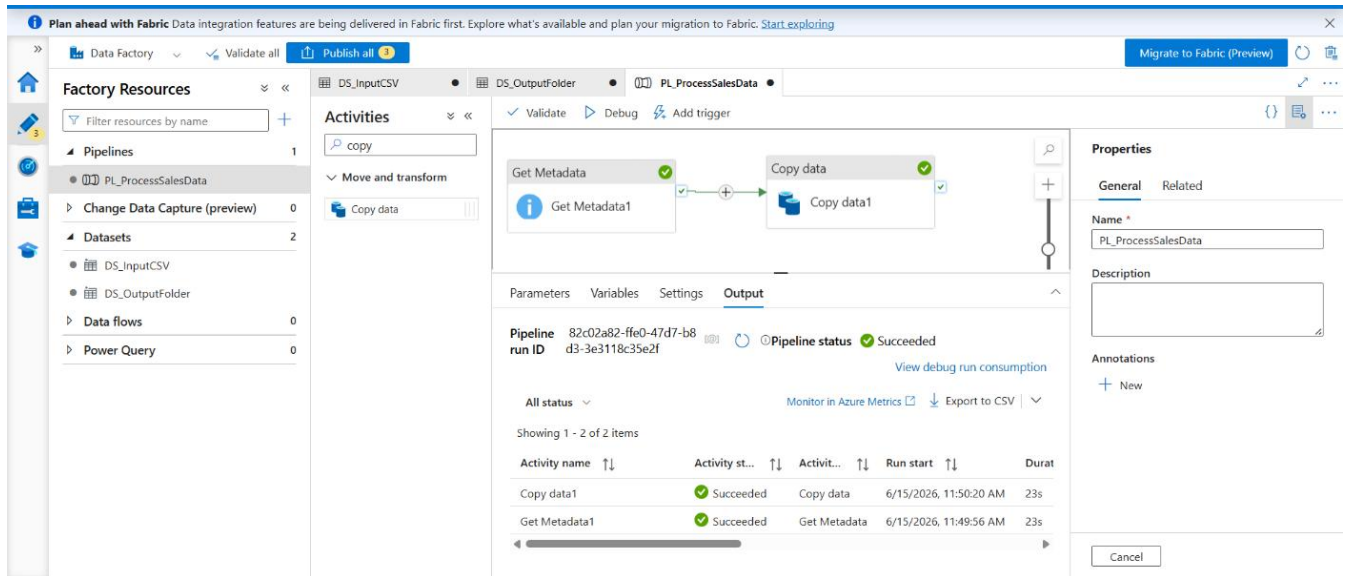
Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



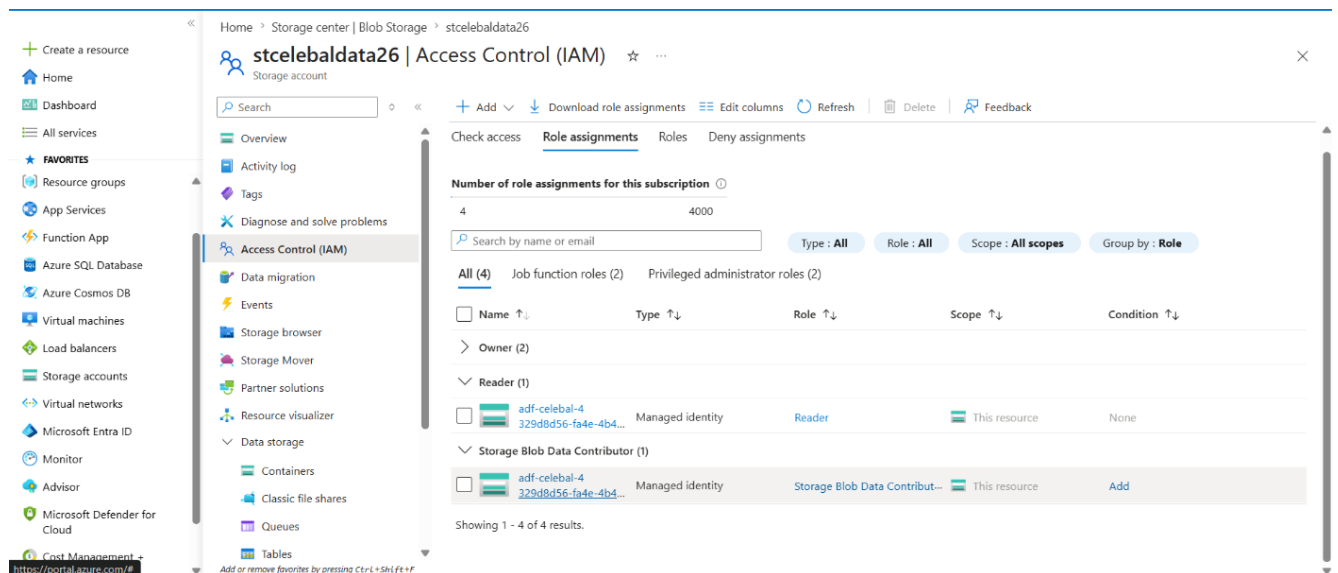
Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 1
 - PL_ProcessSalesData
- Datasets: 2
 - DS_InputCSV
 - DS_OutputFolder
- Data flows: 0
- Power Query: 0

Activities

- copy
- Move and transform
 - Copy data

Get Metadata1 → Copy data1

Parameters Variables Settings **Output**

Pipeline run ID: 82c02a82-ffe0-47d7-b8d3-3e3118c35e2f Pipeline status: Succeeded

View debug run consumption

All status: Monitor in Azure Metrics Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Durat
Copy data1	Succeeded	Copy data	6/15/2026, 11:50:20 AM	23s
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 11:49:56 AM	23s

Properties

General Related

Name: PL_ProcessSalesData

Description

Annotations: + New

Cancel

- Data copied to destination

DS_InputCSV DS_OutputFolder PL_ProcessSalesData

Activities

- copy
- Move and transform
 - Copy data

Get Metadata1 → Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset: DS_OutputFolder

Copy behavior: Select...

Max concurrent connections:

Block size (MB):

Metadata: + New

Quote all text: ☒

Properties

General Related

Name: PL_ProcessSalesData

Description

Annotations: + New

Cancel

- Metadata validated

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>>

Output

Copy to clipboard

```
{
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 7
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```